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SPORT AND EXERCISE PSYCHOLOGY

# Napping behaviour, daytime sleepiness, and arousal in high performance athletes and non-athlete controls

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**ABSTRACT**

Napping offers a strategy to manage sleep, aid recovery and enhance performance in elite sport. However, relatively little research attention has focussed on the natural history of athlete napping or tested the widely held assumption that athlete napping is mainly a consequence of degraded night-time sleep. Within a sample of 158 team ($n = 76$) and individual ($n = 80$) sport athletes, and 82 non-athlete controls, we analysed napping behaviour in relation to sleep quality, daytime sleepiness, and pre-sleep somatic and cognitive arousal. There was no significant association between athlete/non-athlete status and the prevalence, frequency, or duration of naps. Comparisons of athlete nappers and non-nappers found no significant differences in sleep quantity or quality. While nap propensity was significantly related to higher daytime sleepiness, this influence was moderated or augmented by levels of pre-sleep cognitive arousal. For some nappers, those with higher levels of arousal may need to be sleepier than those with lower levels of arousal in order to successfully initiate daytime sleep. Approximately 50% of athletes did not nap. If the benefits of athlete napping are to be fully exploited, the needs of this substantial group for whom napping may be problematic should be recognised and addressed.

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Athletes; sleep; sleepiness; nap; arousal

## Introduction

Experimental and observational studies indicate that appropriately scheduled daytime naps can enhance performance and improve sleep management among elite athletes across a range of elite sports (Bonnar et al., 2018; Botonis et al., 2021; Lastella et al., 2021; Romdhani et al., 2021). However, since access to these benefits is gated by an ability to initiate diurnal sleep, it is possible that advice to nap during the day may not be actionable for many athletes. While clearly influenced by homoeostatic state (see Dinges, 1992), the evidence also suggests that sleepiness is not a sufficient condition for napping, and that nap propensity also reflects trait-like individual differences. Sleep difficulties and sleep quality, for example, do not differentiate groups of nappers from non-nappers (Dinges, 1992, p. 126), and daytime sleep latencies for some elite athletes are independent of subjective sleepiness (Gupta et al., 2021). Individual differences in nap behaviour are also suggested in population studies. Epidemiological surveys repeatedly show that most adults do not habitually nap (Léger et al., 2020; National Sleep Foundation, 2013). Similarly, studies conducted among elite squads of junior rowers (Kölling et al., 2016) and senior athletes from swimming, road cycling, triathlon, mountain biking, race walking, basketball and Australian rules football (Sargent et al., 2014) report nap prevalence rates below 50% during normal training. That napping reflects genotypic traits is strongly suggested by Maire et al. (2014) who report that, following sleep deprivation, allele length of the PERIOD3 (PER3) clock gene is significantly associated with both the experience of sleep debt, and the likelihood of napping, with carriers of the longer PER35/5 allele showing a faster build-up of sleep pressure, and greater diurnal sleep ability (i.e., ability to nap) relative to carriers of the shorter PER34/4 allele. A candidate mechanism here, which shows similarly strong genetic links, is pre-sleep arousal. Using data from the G1219 twin/sibling study, Schneider et al. (2019) showed that overall levels of pre-sleep arousal (as measured by the Pre-Sleep Arousal Scale) are highly correlated with insomnia symptoms, and show significant levels of heritability. Higher levels of trait arousal inhibit sleep onset (see Palagini et al., 2016; Spiegelhalder et al., 2012), helping to explain the apparently paradoxical finding that those with the lowest quality (Perusse et al., 2013; Roehrs et al., 2011) and quantity (Roehrs et al., 2011) of night-time sleep also experience the greatest difficulty initiating and maintaining daytime naps. It is reasonable to hypothesise, therefore, that higher levels of somatic or cognitive arousal may inhibit daytime napping. Such mechanisms may not only contribute to athlete napping patterns in general, but may also help to explain sleep differences reported for individual and team sports. Erlacher et al. (2011), for example, found poorer sleep quality among athletes from athletics, swimming, dancing, figure skating, karate and tennis compared to those from basketball and soccer, while Lastella et al. (2015) reported higher nap propensities among athletes from mountain biking, racewalking, swimming, cycling and triathlon relative to those from Australian rules football, basketball, football and rugby union. Athlete sleep and napping patterns could also be informed by comparisons with non-athletes, where differences would help identify factors, and possibly

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phenotypes, prevalent in elite sport. Studies contrasting athlete and non-athlete sleep, however, are lacking. As a result, the consensus view that " . . . Elite athletes are particularly susceptible to sleep inadequacies" (Walsh et al., 2021), while implicitly comparative, is empirically poorly supported. Few studies have directly compared athlete sleep quantity and quality with those found in the general population, and no study has yet compared individual differences which may impact the napping behaviour of both athletes and non-athletes.

If strategic napping is to be optimally deployed in elite sport, research is needed which helps to identify and overcome barriers to implementation. The present study, therefore, was designed: 1) to assess and compare athlete and non-athlete napping patterns in relation to homoeostatic sleep pressure and pre-sleep arousal; and 2) to contrast the sleep and napping patterns of team and individual sport athletes, again in relation to sleep homoeostasis and pre-sleep arousal. Analyses addressed five research questions:

(1) Is the prevalence of napping significantly higher among athletes when compared with healthy non-athletes?
(2) Do athlete sleep profiles differ significantly from those of healthy non-athletes?
(3) Do sleep profiles differ significantly between athletes who do, and do not habitually nap?
(4) In models adjusted for homoeostatic sleep pressure, is athlete napping significantly associated with lower levels of pre-sleep arousal?
(5) Do the sleep, napping and arousal profiles of individual sport athletes differ significantly from those of team sport athletes?

## Method

### Design and participants

The study was designed as a cross-sectional online survey using the software Qualtrics. The study was approved by Ethics Approvals (Human Participants) Sub-Committee. For athletes, study information was distributed to sports organisations via social media or direct contact with teams and coaches. Non-athletes were recruited through local advertisements and social media. Study literature emphasised sleep, activity and health (but did not specifically emphasise napping) and provided the link to the survey website which provided study information, a screening questionnaire, informed consent procedures and, for those wishing to progress, the survey questionnaire. Recruitment continued until minimum sample sizes were exceeded. All participants were offered personalised sleep profiles on completion of data collection. Of 262 participants who completed the online questionnaires, 22 did not meet the study inclusion criteria and were removed. The final sample of 240 comprised 158 athletes and 82 non-athletes. Athletes reported a mean age of 23.8 year, with a prevalence of female athletes of 64.6%. Athletes competed regionally (23.4%), nationally (36.1%), internationally (29.1%) or professionally (11.4%). Within the competitive/training cycle, 63 (39.9%) athletes were in pre-season training, 68 (43%) were in the competitive season, while 27 (17.1%) reported being in off-season. All participants were resident in the United Kingdom.

### Inclusion criteria

For athletes, inclusion criteria were: at least semi-elite (using the taxonomy proposed by Swann et al., 2015), non-smoking and aged ≥18 years; for non-athletes: aged ≥18 years; non-smoking; not engaged in competitive sport, non-obese; and engaged in moderate to vigorous physical activity for at least 150 minutes/week (as recommended by the World Health Organization (WHO), 2010).

### Outcome measures

Nap categories were based on responses to the single item: "Do you nap during the day?" Those responding "yes" were categorised as nappers; those responding "no" were categorised as non-nappers. Nappers were then asked: "How frequently do you nap" (Less than 1/week/More than 1/week); "For how long do you sleep when you nap" (<1 hour/>1 hour); "Do you find it easy to fall asleep when you take a nap?" (Mostly yes/Mostly no); and "Why do you usually nap?" (All participants: to relieve feelings of sleepiness; because I have spare time; to compensate for poor sleep at night; to avoid feeling sleep later in the day; other. For athletes the additional option "to aid recovery" was added. Non-nappers were asked "What is your reason/are your reasons for not napping during the day?" (It negatively affects my subsequent sleep quality at night; it makes me feel worse for the rest of the day; I have attempted it previously and I cannot fall asleep; I feel napping would attract disapproval; I do not feel sleepy in the day; I see no benefit in it; other).

The questionnaire included the following scales:

#### Pittsburgh sleep quality index: PSQI (Buysse et al., 1989)

The 19-item PSQI is designed to measure sleep disturbance and quality over the previous 1 month. Global scores range from 0–21, with higher scores indicating poorer sleep quality. PSQI component scores include subjective total sleep time (TST).

#### Pre-sleep arousal scale: PSAS (Nicassio et al., 1985)

The PSAS comprises two 8-item subscales assessing symptoms of somatic (e.g., intense breathing, increased heart rate) and cognitive (e.g., worrying, mental alertness) arousal when in bed attempting to sleep. The intensity of experience associated with each symptom is rated on a 5-point scale (1 = "not at all"; 5 = "extremely"). Higher scores indicate higher levels of pre-sleep arousal.

#### Epworth sleepiness scale: ESS (Johns, 1991)

The ESS assesses daytime sleepiness by asking respondents to rate the possibility of "dozing off" in relation to eight scenarios. Ratings range from 0 (never doze off) to 3 (high chance of dozing) with higher scores indicating greater daytime sleepiness. Designed to identify the magnitude of daytime

sleepiness, analyses indicate that reliability is greatest when the ESS is used as a categorical measure (see Walker et al., 2020).

### *Generalised anxiety disorder 7-item scale: GAD-7 (Spitzer et al., 2006)*

The GAD-7 assesses anxiety symptomatology over the previous two weeks (*F*eeling nervous, anxious, etc). Scores range from 0–21, with higher scores indicating higher levels of anxiety.

### *Ford insomnia response to stress test: FIRST (Drake et al., 2004)*

Designed to quantify vulnerability to insomnia ("sleep reactivity"), the FIRST presents nine scenarios which are rated for degree of situational sleep difficulty on a 1–4 scale. The nine items are additive, with higher scores indicating greater sleep reactivity.

### *Morningness-eveningness questionnaire: rMEQ (Adan & Almirall, 1991)*

The rMEQ uses five items from Horne and Östberg (1976) MEQ, to identify the chronotypes: "Definitely Morning Type", "Moderately Morning Type", "Neither Type", "Moderately Evening Type", and "Definitely Evening Type".

### *Sample sizes*

Sample sizes were estimated using G*Power (Faul et al., 2007). The design required sufficient participants to support combined athlete/non-athlete analyses in ANCOVA models, and sub-group comparisons of athlete nappers versus athlete non-nappers. With alpha at 0.05, beta at 80%, numerator df = 7, and allowing for two covariates, models required a minimum total sample size of 237. For logistic regression analyses of the athlete sample alone, a 20% difference in napping/non-napping (65% v 45%) for the higher/lower levels of ESS Sleepiness respectively, with alpha at 0.05 and beta at 80%, required a minimum total sample size of 154.

### *Data analysis*

In preliminary bivariate analyses, non-athletes showed a significantly higher average age than athletes (23.4 years v 30.4 years; $t = 5.72$; $p<0.001$). For continuous variables, therefore, athletes and non-athletes were compared in factorial analysis of covariance (ANCOVA) models with age as a covariate, the target variable of interest as dependent, and participant status (athlete/non-athlete and napper/non-napper) as grouping factors. To address research question 4, the grouping factor propensity for sleepiness was added to this model. Prior to modelling, the assumption of homogeneity of regression slopes was assessed for all outcomes by testing group by covariate interaction terms; none achieved significance. For comparisons within the athlete group alone, factorial ANOVA models were fitted with napper/non-napper, and team sport/individual sport participation as grouping factors. All between group mean differences are reported as main effects. Significant interactions were evaluated by comparing levels within factors using Bonferroni adjusted post-hoc tests. Categorical outcomes were analysed using chi-square, or binary logistic regression. Logistic regressions are reported as odds ratios with 95% confidence intervals, and the significance of the Wald $X^2$ statistic. For rMEQ scores the "Definitely Evening" and "Definitely Morning" responses were collapsed into the "Moderately Evening and Moderately Morning" categories respectively because of low cell frequencies at the extremes. All analyses were performed using IBM SPSS Statistics Package 27.0 (IBM, Armonk, NY, USA). Significance levels were set at 0.05 throughout.

## Results

### *Athlete versus non-athlete comparisons*

The final sample comprised 158 athletes and 82 non-athletes (Table 1). Among athletes, the most frequently represented sports were athletics (28%), rugby union (27%), cycling (14%), cricket (10%), football (3%), swimming (3%) field hockey (2%) and weightlifting (2%). The remaining 11% included athletes from rowing, netball, basketball, squash, boxing, skiing, gymnastics, lacrosse, sailing and martial arts.

Cronbach's alpha coefficients for each of the administered scales were as follows: PSQI .64; PSAS-S .80; PSAS-C .91; ESS .80; GAD-7 .90; FIRST .80; and rMEQ .69. A median split of ESS scores (7) was used to divide the whole sample into "higher

**Table 1.** Comparison of athlete and non-athlete demographic, sleep and psychometric characteristics.

| Variable | Athletes ($N = 158$) | Non-athletes ($N = 82$) | p value/OR (95% CI) |
|---|---|---|---|
| Mean age (SE) | 23.8 (0.59) | 30.4 (1.14) | $p<0.001$[a] |
| Gender: N (%) female | 102 (64.6) | 50 (61.7) | $p=.46$[b] |
| Nap prevalence: N (%) | 81 (51.3) | 30 (36.6) | 0.63 (0.35–1.13)[c] |
| Nap frequency > 1/week: N (%)[d] | 53 (65.4) | 23 (76.7) | 1.72 (0.64–4.63)[c] |
| Typical nap duration: N (%. <1 hour) | 33 (40.7) | 10 (33.3) | 0.95 (0.90–1.00)[c] |
| Easy to nap: N (% "yes")[d] | 64 (79.0) | 25 (83.3) | 0.75 (0.24–2.31)[c] |
| PSQI | | | |
| PSQI Global score: mean (SE) | 6.2 (0.23) | 6.8 (0.35) | $p=.13$[e] |
| PSQI Total sleep time (minutes): mean (SE) | 456.2 (5.73) | 418.8 (8.33) | $p<0.001$[e] |
| PSAS-S score: mean (SE) | 12.7 (0.38) | 13.2 (0.56) | $p=.44$[e] |
| PSAS-C score: mean (SE) | 19.9 (0.58) | 22.2 (0.84) | $p<0.03$[e] |
| ESS Higher sleepiness: N (%) | 71 (44.9) | 45 (54.9) | 1.86 (1.03–3.34)[c] |
| FIRST score: mean (SE) | 20.8 (0.46) | 22.2 (0.66) | $p=.09$ |
| GAD-7 score: mean (SE) | 5.5 (0.38) | 7.3 (0.55) | $p<0.01$[e] |

**Abbreviations** OR: odds ratio; CI: confidence interval; PSQI: Pittsburgh Sleep Quality Index; PSAS: Pre-sleep Arousal Scale (S = somatic arousal; C = cognitive arousal); ESS: Epworth Sleepiness Scale; GAD-7: Generalised Anxiety Disorder scale.

[a]independent sample t-tests; [b]significance of Pearson $X^2$; [c]odds ratio and 95% confidence interval from binary logistic regression model adjusted for age; [d]nappers only; [e]significance of main effect F-ratio from univariate ANCOVA adjusted for age (adjusted marginal mean values reported).

sleepiness" (≤7) and "lower sleepiness" (>7) sub-groups. This division was validated against total sleep time (TST) scores from the PSQI, which showed significantly longer TSTs for the lower sleepiness grouping (total sample age adjusted F(1,239) = 12.36, $p < 0.001$; mean difference = 32.0 minutes. Athletes alone: $t = 3.27$, $p < 0.001$; mean difference = 30.0 minutes)

Sleep and sleep-related profiles of athletes and non-athletes are compared in Table 1. Age adjusted logistic regression models found no significant differences between athletes and non-athletes in the prevalence of habitual napping (OR = 0.63, 95% CI: 0.35–1.13, $p = .12$), nap frequencies of more than 1/week (OR = 1.72, 95% CI: 0.64–4.63, $p = .28$), typical nap durations of > 1 hour (OR = 0.95, 95% CI: 0.90–1.00, $p = .23$), or finding it generally "easy" to initiate a nap (OR = 0.75, 95% CI: 0.24–2.31, $p = .61$). Age controlled athlete versus non-athlete ANCOVA comparisons also showed no significant differences in PSQI global scores ($F(1,239) = 2.56$, $p = .13$), PSAS-S scores ($F(1,239) = 0.60$, $p = .44$, or FIRST scores ($F(1,239) = 2.93$, $p = .09$). However, athletes showed significantly longer mean total sleep times ($F(1,239) = 13.03$, $p<0.001$), significantly lower PSAS-C scores ($F(1,239) = 4.73$, $p = .03$), and GAD-7 anxiety scores ($F(1,239) = 6.86$, $p = .01$), and were significantly less likely to score in the ESS "high sleepiness range" (OR = 1.86, 95% CI: 1.03–3.34, $p < 0.04$), relative to non-athletes. None of the ANCOVA models reported in Table 1 showed a significant athlete status by napper status interaction effect.

In regard to research question 4 ("In models adjusted for homeostatic sleep pressure, is athlete napping significantly associated with lower levels of pre-sleep arousal?) the ANCOVA model showed that PSAS-C scores were significantly lower among those nappers who also reported lower levels of sleepiness. This significant nap status by sleepiness interaction (F (1,231) 8.12, $p = 0.005$) is shown in Figure 1. Paired comparisons showed that the nappers in the lower sleepiness sub-group reported significantly lower levels of pre-sleep cognitive arousal than both non-nappers (mean difference = 4.2, $p = .005$; Figure 1) and nappers in the higher sleepiness sub-group (mean score difference = 4.1, $p < 0.03$; Figure 1). These effects were independent of athlete/non-athlete status.

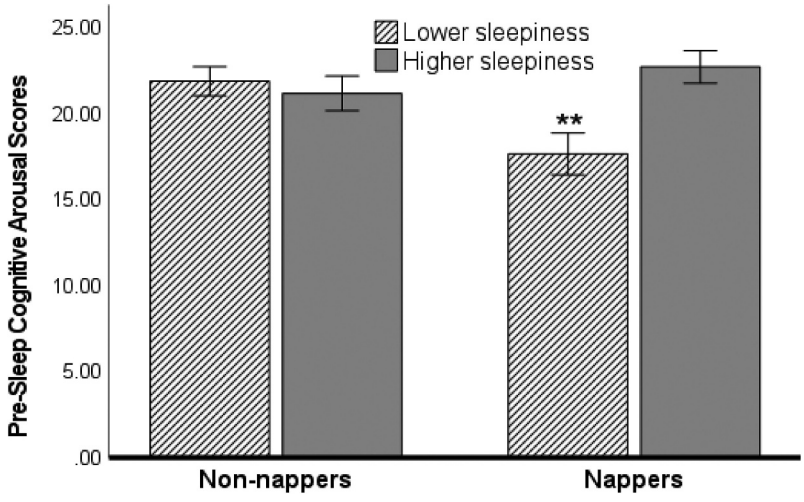


Figure 1. Levels of pre-sleep cognitive arousal by nap status (napper/non-napper) and sleepiness (lower/higher) in a combined sample of athletes ($n = 158$) and non-athletes ($n = 82$). Nap status x sleepiness interaction: ($F$ (1,231) 8.12, $p = .005$). The athlete v non-athlete main effect was not significant. **differs significantly from all other categories at $p<0.05$

Table 2. Comparison of demographic and sleep characteristics of 158 high-performance athletes categorised as nappers and non-nappers.

| Variable | | Nappers | Non-nappers | p value |
|---|---|---|---|---|
| N (%) | | 81 (51.3%) | 77 (48.7%) | |
| Mean Age (SE) | | 23.6 (0.80) | 24.0 (0.86) | $p$=.69[a] |
| Gender: N (%) female | | 49 (60.5%) | 53 (68.8%) | $p$=.25[b] |
| PSQI global score: Mean (SE) | | 6.3 (0.32) | 6.1 (0.30) | $p$=.68[c] |
| PSQI total sleep time (min): mean (SE) | | 459.6 (7.52) | 455.3 (7.66) | $p$=.69[c] |
| PSAS-S score: mean (SE) | | 12.8 (0.55) | 12.8 (0.53) | $p$=.88[c] |
| PSAS-C score: mean (SE) | | 19.6 (0.71) | 21.2 (0.83) | $p$=.14[c] |
| ESS Higher sleepiness: N (%) | | 49 (60.5) | 22 (28.6) | $p$<0.001[b] |
| FIRST score: mean (SE) | | 20.7 (0.60) | 21.3 (0.64) | $p$=.47[c] |
| GAD-7 score: mean (SE) | | 5.4 (0.54) | 5.5 (0.55) | $p$=.92[c] |
| rMEQ | Definitely/Moderately Evening Type: N (%) | 15 (18.5) | 14 (18.2) | $p$=.32[b] |
| | Neither Type | 50 (61.7) | 40 (51.9) | |
| | Definitely/Moderately Morning Type: N (%) | 16 (19.8) | 23 (29.9) | |

**Abbreviations** PSQI: Pittsburgh Sleep Quality Index; PSAS: Pre-sleep Arousal Scale (S = somatic arousal; C = cognitive arousal); ESS: Epworth Sleepiness Scale; GAD-7: Generalised Anxiety Disorder scale; rMEQ: Reduced Morningness – Eveningness Questionnaire.
[a]independent sample t-tests; [b]significance of Pearson X2; [c]significance of main effect F-ratio.

Table 3. Binary logistic regression analyses of factors associated with athlete napping behaviour.

| Variables in Model | Napper | Non-napper | OR (95% CI) |
|---|---|---|---|
| ESS score: N (%) higher sleepiness | 49 (60.5) | 22 (28,6) | 4.91 (2.38–10.13) |
| PSAS-C score: mean (SE) | 19.6 (6.4) | 21.2 (7.3) | 0.95 (0.89–0.99) |
| Total Sleep Time: mean (SE) | 542.6 (8.7) | 529.5 (7.9) | 1.0 (1.0–1.0) |

**Abbreviations** OR: odds ratio; CI: confidence interval; ESS: Epworth Sleepiness Scale; PSAS-C: Pre-sleep Arousal Scale (cognitive arousal).

[a]binary logistic regression model with nap-status as dependent and ESS, PSAS-C, and total sleep time as covariates.

### *Athlete napper/athlete non-napper comparisons*

Sleep and sleep related characteristics of athlete nappers and non-nappers are compared in Table 2. Nap prevalence was 51.3% (95% CI: 43.5–58.9), with napper/non-napper status showing no significant association with age ($t = 0.40$, $p = .69$) or gender ($X^2$ 1.33, df = 1, $p$=.25). Principal reasons offered for napping were: to relieve feelings of sleepiness ($n = 76$; 93.8%); to compensate for poor sleep at night ($n = 65$; 80.2%); and to aid recovery ($n = 55$; 67.9%). Principal reasons given for not napping were: negative impact on night-time sleep ($n = 48$; 62.3%); and previously experienced difficulty initiating naps ($n = 44$; 57%).

Nappers and non-nappers showed no significant differences in PSQI global sleep quality scores ($F$ (1, 157) 0.18, $p = .68$), total sleep time ($F(1,157) = 0.16$, $p = .69$), somatic ($F(1,157) = 0.02$, $p = .90$) and cognitive ($F(1,157) = 2.21$, $p$=.14) pre-sleep arousal, or GAD-7 anxiety scores ($F(1, 157) = 0.01$, $p = .92$). Nap status also showed no significant association with chronotype ($X^2$ 2.30, df = 2, $p$=.32), but did show a highly significant association with ESS daytime sleepiness, with a greater proportion of nappers reporting "higher sleepiness" ($X^2$ 16.26, df = 1, $p$<0.001; Table 2).

To assess the influence of cognitive arousal on nap propensity after adjustment for homoeostatic sleep pressure, and further address research question 4, PSAS-C scores, total sleep time scores and ESS higher sleepiness/lower sleepiness categories were entered in a binary logistic regression models with napper/non-napper as the dependent variable. In a preliminary analysis, the interaction term PSAS-C by ESS was non-significant and thus removed from the model. The final model was significant ($X^2 = 22.77$, df = 4, $p$<0.001), explaining 17.9% of variance in napping behaviour (Nagelkerke $R^2$). Results are shown in Table 3. Membership of the "higher sleepiness" category was significantly associated with an almost 5-fold increased probability of napping (OR = 4.91; 95% CI = 2.38–10.13), while each increase of 1 point on the PSAS-C scale was significantly and independently associated with a 5% decrease in the probability of napping (OR = 0.95; 95% CI = 0.89–0.99). Total sleep time was not significantly associated with napping behaviour in this model.

### *Napping in team and individual sport athlete*

Comparisons of team and individual sport athletes are shown in Table 4. Main effects from the ANOVA models showed no significant differences in PSQI scores ($F(1,157) = 0.98$, $p$=.33), total sleep time ($F(1,158) = 0.89$, $p$=.35), or GAD-7 scores ($F(1,157) = 2.55$, $p$=.11). PSQI scores did, however, show a significant nap status by sport type interaction ($F(1,157) = 5.909$, $p$=.02), with individual sport nappers showing significantly lower sleep quality than team sport nappers (mean difference 1.25, $p$=.043). Individual sport athletes also showed significantly higher levels of PSAS somatic ($F(1,157) = 13.25$, $p$<0.001) and cognitive ($F(1,157) = 6.59$, $p$=.01) arousal, and significantly greater FIRST sleep reactivity scores ($F(1,157) = 11.27$, $p$=..001).

Categorical outcomes showed no significant associations between participation in team/individual sports and the prevalence of habitual napping ($X^2 = 0.03$,df = 1, $p$=.87), nap frequencies of more than 1/week ($X^2 = 0.01$, df = 1, $p$=.94), typical nap durations of > 1 hour ($X^2 = 0.02$, df = 1, $p$=.89), finding it generally "easy" to initiate a nap ($X^2 = 0.58$, df = 1, $p$=.45), or higher levels of ESS daytime sleepiness ($X^2 = 1.33$, df-1, $p$=.25).

## Discussion

The present study set out to investigate athlete napping in relation to homoeostatic sleep pressure and pre-sleep arousal. Given that the initial invitation to participate in this study did not emphasise napping, and that napping items were embedded among questions addressing sleep, health, exercise and psychological status, it is unlikely that the data reflect a systematic recruitment bias towards either nappers or non-nappers. Collectively, the results support three broad conclusions. First, that prevalence rates of athlete napping show a range similar to that reported for the general population. Second, that among high-performance athletes, daytime nap propensity is inadequately characterised as a simple function of night-time sleep quality or quantity. And third, that among nappers, the influence of homoeostatic sleep pressure (as indexed by reported daytime sleepiness) is moderated or augmented by levels of pre-sleep cognitive arousal. Support for these conclusions is drawn from comparisons of: athletes and non-athletes; athlete nappers and athlete non-nappers; and team and individual sport athletes.

Comparisons of athletes and non-athletes (Table 1) offer little support for the general proposition that athlete sleep profiles are characterised by poorer quality or restricted sleep. Athletes and non-athletes showed no significant differences in global sleep quality (PSQI scores), or the prevalence, frequency, or duration of reported daytime naps. For both groups, nap prevalence rates accord with the values (35%-51%) reported for adult napping generally (National Sleep Foundation, 2013; Léger et al., 2020). However, athletes did show significantly longer total sleep times, and significantly lower levels of daytime sleepiness, anxiety and pre-sleep cognitive arousal relative to non-athletes, all suggestive of lower sleep challenge. Differences between the present (Table 1) findings and

previously reported survey comparisons of athlete/non-athlete sleep quality are likely due to control group selection. Demirel (2016), for example, reported significantly higher PSQI scores for athletes when compared with "sedentary" non-athletes, while Bender et al. (2019) reported significantly lower PSQI scores for athletes when compared with a sample of screened non-athlete "good sleepers". In contrast, our selection criteria aimed to recruit control participants who differed from athletes in terms of engagement in competitive sport, but who were nevertheless active and healthy.

Results from napper versus non-napper comparisons (Table 2) also challenge the conclusion that athlete napping mainly reflects the adequacy of night-time sleep. That nappers showed no significant differences from non-nappers in sleep quality (PSQI scores), sleep quantity, chronotype, and anxiety, but nevertheless reported significantly higher levels of daytime sleepiness, suggests the influence of additional, intervening factors in the pathway from night-time sleep, through sleepiness, to daytime napping. A greater sleep requirement among nappers, for example, might explain why napper/non-napper similarities in total sleep at night could result in elevated napper sleep need in the day. However, such an explanation would less plausibly account for the comparable subjective sleep quality scores or chronotype distributions reported here. On the other hand, evidence from both the factorial model fitted to the combined athlete/non-athlete data (Figure 1), and the logistic regression analysis of athlete nappers/non-nappers (Table 3), confirms the influence of the arousal construct on daytime nap propensity. Independent of athlete/non-athlete status, cognitive arousal demonstrated a significant moderating effect on sleepiness (Figure 1). Nappers in the lower sleepiness sub-group showed significantly lower PSAS-C scores than both higher sleepiness nappers and lower sleepiness non-nappers. A reasonable interpretation of this finding is that athlete and non-athlete nappers with lower levels of pre-sleep cognitive arousal needed less "sleepiness" to nap successfully than those with higher levels of pre-sleep cognitive arousal. This interpretation is in line with earlier findings suggesting that cognitive hyperarousal "... predominates over mental fatigue" (Perusse et al., 2013, p. 1279). Additionally, the results in Table 3 show that, independent of sleepiness and night-time sleep duration, cognitive arousal represents a modest but significant influence on athlete nap probability, varying the odds of napping by 5% with each point on the PSAS-C scale.

Comparisons between individual and team sport athletes also reflect an influential role for the arousal construct in athlete sleep and napping behaviour. Individual and team sport membership showed no significant association with sleepiness categories or nap status in bivariate comparisons. Nevertheless, the highly significant differences in PSAS-S, PSAS-C and FIRST shown in Table 4 clearly indicate higher levels of sleep vulnerability and reactivity among individual sports athletes. Evidence provided by the analyses of global PSQI scores, however, indicates that this greater vulnerability is differentially linked to napping behaviour. While the PSQI ANOVA showed no team sport versus individual sport main effect (see Table 4), it did show a significant nap status by sport type interaction, with individual sport nappers reporting significantly lower sleep quality than both team sport nappers and individual sport non-nappers. For individual sport athletes, then, poor sleepers are concentrated among nappers. This finding could help to explain earlier, equivocal results from individual and team sport comparisons. Erlacher et al. (2011) found a significantly higher frequency of reported sleep problems among individual sport athletes, while Lastella et al. (2015) report no overall difference in subjective sleep quality. These studies report only individual/team group comparisons, and did not examine sleep quality scores in relation to individual napping behaviours.

**Table 4.** Comparison of demographic and sleep characteristics of 158 high-performance team and individual sport athletes.

| Variable | | Team sports | Individual sports | p value |
|---|---|---|---|---|
| N (%) | | 76 (48.7%) | 80 (51.3%) | |
| Mean Age (SD) | | 24.3 (0.62) | 23.3 (0.98) | $p$=.42[a] |
| Gender: N (%) female | | 51 (67.1) | 50 (62.5) | $p$=.55[b] |
| Nap prevalence: N (%) | | 40 (51.9) | 41 (50.6) | $p$=.87[b] |
| Nap frequency > 1/week: N (%)[c] | | 14 (50.0) | 14 (50.0) | $p$=.94[b] |
| Typical nap duration: N (%. <1 hour)[c] | | 16 (48.5) | 17 (51.5) | $p$=.89[b] |
| Easy to nap: N (% "yes")[c] | | 33 (51.6) | 31 (48,4) | $p$=.58[b] |
| PSQI global score: Mean (SE) | | 6.0 (0.31) | 6.4 (0.31) | $p$=.33[d] |
| PSQI total sleep time (min): mean (SE) | | 462.6 (7.71) | 452.4 (7.52) | $p$=.35[d] |
| PSAS-S score: mean (SE) | | 11.4 (0.53) | 14.1 (0.51) | $p$<0.001[d] |
| PSAS-C score: mean (SE) | | 19.0 (0.77) | 21.7 (0.75) | $p$=.01[d] |
| ESS Higher sleepiness: N (%) | | 31 (40.3) | 40 (49.4) | $p$=.25[b] |
| FIRST score: mean (SE) | | 19.5 (0.54) | 22.4 (0.64) | $p$<0.001[d] |
| GAD-7 score: mean (SE) | | 5.1 (0.53) | 6.3 (0.52) | $p$=.11[d] |
| rMEQ | Definitely/Moderately Evening Type: N (%) | 18 (23.4) | 11 (13.6) | $p$=.32[b] |
| | Neither Type | 46 (59.7) | 44 (54.3) | |
| | Definitely/Moderately Morning Type: N (%) | 13 (16.9) | 26 (32.1) | |

**Abbreviations** PSQI: Pittsburgh Sleep Quality Index; PSAS: Pre-sleep Arousal Scale (S = somatic arousal; C = cognitive arousal); ESS: Epworth Sleepiness Scale; GAD-7: Generalised Anxiety Disorder scale; rMEQ: Reduced Morningness – Eveningness Questionnaire.
[a]independent sample t-tests; [b]significance of Pearson $X^2$; [c]nappers only.
[d]significance of main effect F-ratio.

It should be emphasised that the survey findings reported here are not definitive, but provide a strong empirical basis for experimental studies of the *night-time sleep → daytime sleepiness → napping*' pathway in elite sport. The relationship between arousal and sleepiness merits particular attention. Figure 1 indicates that lower levels of pre-sleep cognitive arousal can diminish the contribution of daytime sleepiness to successful napping. This, in turn, suggests that lower cognitive arousal could contribute to the athlete "sleepability" construct reported by Gupta et al. (2021), who found that elite athlete sleep latencies for afternoon nap opportunities, though shorter than those for non-athletes, were independent of experienced sleepiness. A further consideration is the measurement of the daytime "sleepiness" construct itself in elite sport. Marques et al. (2019), for example, draw a distinction between sleepiness propensity (the expressed likelihood of falling asleep as captured by the ESS), and sleepiness perception (the subjective experience of sleepiness). Contrasting results from the ESS and the 4-item DSPS-4 (a measure of sleepiness perception), Marques et al. (2019) found that PSAS-C scores correlated positively and significantly with DSPS-4 scores, but showed no significant correlation with ESS scores. Viewed in the light of the present study outcomes, such findings suggest that future studies exploring athlete napping behaviour should include measures of both sleepiness propensity and perception.

No previous study has considered why athletes do not nap. Non-napping was reported by 48.7% of athletes, and linked to lower levels of experienced daytime sleepiness and higher levels of pre-sleep cognitive arousal. These findings accord with the reasons offered by athletes for not napping, with 57% of non-nappers reporting that they experienced difficulty initiating naps. If elite sport adopts daytime napping as a strategy to enhance performance or manage recovery (see Lastella et al., 2021), then the needs of this substantial group of athletes for whom napping may be problematic should be recognised and addressed.

Three important limitations should be acknowledged when interpreting the present results. First, the large number of comparisons conducted would have increased the risk of Type 1 error. In mitigation, all post hoc comparisons were Bonferroni adjusted, the overall pattern of results was consistent and interpretable, strongly suggesting the operation of "true" effects, and the high levels of statistical significance that emerged in many of the models all provide additional confidence in reported outcomes.

Second, the recruitment strategy was designed to deliver athlete and control groups which met the requirements for analysis, rather than demographically representative random samples. In the event, while the athlete group provided a good representation of sports, women appear to be over-represented. As indexed by participation in winter and summer Olympics, women represent 45–48% of elite athletes (International Olympic Committee, 2021), while representing 61.7% of the present athlete sample. These differences in gender composition, however, are unlikely to have seriously distorted the sleep and nap profiles reported here. In detailed comparisons of the sleep of male and female collegiate athletes Carter et al. (2020) found little difference in the prevalence of napping (49% males v 46% females), and no significant differences in nap length, total sleep time, ESS scores or PSQI scores. Nevertheless, caution in extrapolating the prevalence rates reported here would be appropriate.

Third, finding appropriate control groups for high performance athletes was challenging. In order to match for health and physical fitness, the present study employed WHO activity criteria. It is clear from the sample, however, that age and lifestyle could have been more closely matched to the athlete sample. While age was controlled in the analytical models, occupational and family dynamics were not. The greater levels of sleepiness reported by non-athletes, then, may be a reflection of situational factors unrelated to age or fitness. That said, there is no consensus on which attributes should be matched to achieve appropriate controls for epidemiological studies of high-performance athletes. In the present study, health and activity status were prioritised on both logical and practical grounds.

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